

1. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8\}$ be the universal set, and $A = \{1, 2, 3, 4\}$ and $B = \{4, 6, 8\}$ be subsets of U . Find the indicated set, and draw a Venn diagram corresponding to the indicated set.
 - a. $A \cup B$
 - b. $A \cap B$
 - c. A'
 - d. B'
 - e. $A \cap B'$
 - f. $A' \cup B$
 - g. $(A \cap B)'$
 - h. $(A \cup B)'$
2. A college has 1105 students. This semester 320 students are taking a math course, 405 of the students are taking a history class, and 250 of the students taking a math class are not enrolled in a history class.
 - a. Construct a Venn diagram, and write the appropriate numbers in the four regions
 - b. How many of the students are taking math and history?
 - c. How many students are taking history but not math?
 - d. How many students are taking math or history?
 - e. How many students are taking neither math nor history?
3. *Survivor*, *CSI*, and *ER* all belong to the Thursday night prime time lineup. A group of 600 college students were surveyed. 350 students watch *Survivor*, 280 watch *CSI*, and 320 watch *ER*. 50 students watch all three shows. 190 students watch both *Survivor* and *ER*. 180 students watch both *Survivor* and *CSI*. 50 students watch only *ER*.
 - a. Construct a Venn diagram, and write the appropriate numbers in the eight regions.
 - b. How many students watch only *Survivor*?
 - c. How many students watch *CSI* or *Survivor*?
 - d. How many students watch *CSI* or *ER*?
 - e. How many students watch only *CSI*?
4. Find the value of each of the following:
 - a. $4!$
 - b. $7!$
 - c. $0!$
 - d. $\frac{10!}{7!}$
 - e. $\frac{27!}{4!25!}$
 - f. $\frac{1000!}{998!}$
 - g. ${}_7P_2$
 - h. ${}_6C_4$

5. A certain model of pickup truck is available in five exterior colors, three interior colors, and three interior styles. In addition, the transmission can be either manual or automatic, and the truck can have either two-wheel or four-wheel drive. How many different versions of the pickup truck can be ordered?
6. A certain fraternity must elect a president, vice president, and treasurer from its 30 members. How many ways can this be done?
7. The local branch of a national business chain has 27 employees. If they all shake each other's hand at the company picnic, at least how many handshakes occurred?
8. Using a standard deck of 52 cards,
 - a. how many ways can five cards be drawn?
 - b. how many five card hands can be dealt?
9. A three-person committee must be chosen from a group of ten men and twelve women. How many committees are possible if it must consist of the following?
 - a. Three men.
 - b. Three women.
 - c. One woman and two men.
 - d. One man and two women.
 - e. One man and two women, and one of these is selected as a spokesperson of the committee.
10. Each student at State University has a student ID number consisting of five digits (the first digit is nonzero, and the digits can be repeated) followed by two of the letters A , B , C , and D (letters cannot be repeated). How many different student numbers are possible?